Global Environmental Change 91 (2025) 102969

Contents lists available at ScienceDirect

Global Environmental Change

journal homepage: www.elsevier.com/locate/gloenvcha

# Addressing climate inaction as our greatest threat to sustainable development

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ARTICLE INFO

*Keywords:*
Systemic market failure
Sustainable development
Climate inaction
Climate policy
Market system
Industrial policy
Climate action

ABSTRACT

More than 1 degree of global warming has been reached and once projected impacts are now being realized. Despite these impacts and the short timeframe available to avoid further warming, climate inaction remains a major threat to sustainable development. In this article, we bring a renewed focus to the issue of climate inaction. We unpack the systemic market failure that underpins current climate action efforts globally and how by shifting focus to address inaction this could be overcome. We explore how climate policies are inadvertently allowing climate inaction to persist, why this is happening and how to address it. Central to our argument is that climate policies still draw too heavily on a neoclassical development paradigm, rather than reinvigorated industrial policy, resulting in market interventions that fail to address the scale and systemic nature of the climate action challenge. We therefore reorient climate policies towards addressing inaction as a systemic development challenge that demands a stronger role from the government. We conclude by proposing a market systems framework for guiding policymakers to better target the systemic nature of climate inaction and the threat it poses to sustainable development.

## 1. Introduction

This article unpacks dimensions of climate inaction as a systemic market failure and arguably the greatest material risk to global sustainable development. When released in 2006, the Stern Review was the world's first large-scale review of how to deal with the challenge that climate change poses to a national economy. It was unequivocal in its assessment that climate change "is the greatest market failure the world has ever seen", but more ominously and perhaps less understood, that inaction presents "very serious impacts on growth and development"(Stern, 2007). While on the surface these two statements have since been widely accepted and syndicated, the second statement on the challenge and potential impact of inaction is noticeably absent as a core objective of climate policies.

Our general concern is with the absence of inaction as a core objective in climate policies and how this omission enables the systemic nature of inaction to go on unchecked (Hornsey and Fielding, 2020). Our more specific concern is that the absence of inaction is representative of a common underlying assumption that impactful climate action across market-based economies is imminently possible with the right policy, plan, intervention or investment. Such assumptions would be in direct contrast to conventional wisdom on the complexity of influencing market systems and driving economy-wide behavior change in market-based economies (Stern et al., 2022). The potential negative consequences of this assumption are becoming evident and deeply concerning. On an economic level, the International Monetary Fund's (IMF) most recent long-term outlook for the global economy estimated that if the current trajectory of climate inaction continues, world real GDP per capita by 2100 will be approximately 7 percent lower than today (Kahn, 2019). On a human level, the 2023 Lancet Countdown on Health and Climate Change highlights that inaction is resulting in health-threatening high-temperatures that are impacting people today, with a 4.7-fold increase in heat-related deaths predicted by mid-century under current levels of inaction (Romanello, 2023). In terms of historical impact, the climate change attributable cost of extreme weather events between 2000–2019 is estimated to be US$2.86 trillion, of which 63 % is due to human loss of life (Newman and Noy, 2023).

So how do sustainable development and climate action efforts

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https://doi.org/10.1016/j.gloenvcha.2025.102969
Received 20 May 2024; Received in revised form 2 December 2024; Accepted 17 January 2025
Available online 29 January 2025

perform under our current policy frameworks? Unsurprisingly, early evidence on the influence of the Sustainable Development Goals (SDG) identifies that cases of transformative impact on the global economy are rare (Biermann, 2022). In the case of SDG 13 climate action, there is a growing recognition that climate policies often fail to elevate and translate their adaptation and mitigation objectives into broader economic impact and sustainable development (Soergel, 2021). A 2021 analysis of sustainable development in all 193 United Nations (UN) member states led by Jeffrey Sachs summarizes the underperformance well. It concluded that while many countries have climate policies in place and the net benefits on offer are clear, global progress is stagnated and insufficient for meeting 2030 climate and SDG targets (Sachs et al., 2021). This situation exists despite (i) decades of research and practice on developing and implementing adaptation and mitigation strategies, policies and plans (Woodruff and Stults, 2016); (ii) increasing evidence that the costs of inaction outweigh the costs of action (Sanderson and O'Neill, 2020); (iii) increasing clarity on the economic and social benefits of transitioning entire societies and economies towards net zero and climate resilience (United Nations, 2020); (iv) widespread public demand for climate action being at an all-time high (Tollefson, 2021); and (v) growing recognition of the underlying significance of SDG 13 climate action in progressing all other 16 SDGs (Fuso Nerini, 2019).

Setting aside the influence that politics can have on climate inaction (Skovgaard and van Asselt, 2019), there appears to be an increasing disconnect between the climate policies seeking to drive climate action, the need for climate action to deliver sustainable development and the very real persistence of climate inaction across market-based economies. Stern's assessment has arguably come true, with climate inaction now being a serious threat to global sustainable development. With time rapidly running out to deliver climate action and set a sustainable development pathway, policymakers and scholars need to be resolute with their focus and attention on the issue of persistent climate inaction.

In this article, we refocus the policy challenge of climate action on the urgent need to address climate inaction as a systemic market failure. First, we clarify the nature of the systemic market failure causing persistent climate inaction by explaining the economic dynamics at play. We then identify four issues with climate policies that explain why they have failed to identify, consider and address climate inaction as a systemic market failure and major development challenge. Based on this problem setting we reframe climate policies towards addressing climate inaction as a development challenge. Finally, we provide a simple market systems development framework and four key policy principles that policymakers can consider in targeting the threat that persistent climate inaction poses to sustainable development.

Our paper defines climate inaction as the prevalence of insufficient progress on addressing human-induced climate change and its impacts, despite climate policy commitments set under the UN Framework Convention on Climate Change (UNFCCC) to reduce greenhouse gas emissions and adapt to climate change impacts. We define a systemic market failure as a chronic market condition where existing market systems are not enabling market actors to prioritise sufficient action and investment towards addressing system-wide threats like climate change. We recognise the ambition of sustainable development as seeking to meet the needs of the present, without compromising the ability of future generations to meet their own needs. Additionally, we recognise the intrinsic link between other aspects of sustainable development and effective climate action, based on the UN SDG 13 Climate Action – Take urgent action to combat climate change and its impacts.

## 2. Unpacking climate inaction as a systemic market failure

Inadequate climate action highlights two economic issues facing sustainable development. The first is market-specific issues that result in a market failure, for example, poor carbon market assurance. In this case, interventions in existing market systems that attempt to strengthen specific market conditions such as regulation, are appropriate contributions to sustainable development (Andrew, 2008). The second recognizes more complex systemic issues, where market failure conditions underlie all aspects of the economy and society. We identify this as a systemic market failure and a major factor leading to climate inaction. In this case, standard market-based interventions are insufficient for realizing sustainable development and much larger interventions are needed to reorient economies and market systems (Bromley, 2007).

Fig. 1 is a representation of non-state actor (the private sector, civil society and individuals) behaviour under current market systems and conditions. It highlights that they continue to place high priority on their own narrow market interests (Hsu et al., 2020), despite the common threat to these interests posed by climate change. This likely occurs because non-state actors are left to their own devices within existing markets, with competing priorities, limited capacity, minimal resources and no platform for collective action (Ray, 1998; Jackson et al., 2021). Importantly, it further highlights that non-state actors are not the driving cause of climate inaction, but may be constrained in taking effective climate action by systemic market failure (Bäckstrand et al., 2017).

Under these market conditions, non-state actors experience constraints to engaging in climate action within the context of their priorities and collective action. For civil society, this would likely mean where possible advocating for and advancing civil conditions that have a direct outcome for community members (Derman, 2014). For the private sector, this likely would mean considering climate change in the development and delivery of products and services and acting within their commercial context and as required by regulation (Linnenluecke et al., 2013). For individuals, this would likely mean changing consumption preferences where possible and without significantly impacting lifestyle or livelihood outcomes (Thøgersen, 2021).

The persistence of a systemic market failure on climate action sends a very clear signal to the market that climate inaction is not an issue worthy of government intervention, undervaluing the important need for transformative climate action in the economy. Such a signal likely has a significant impact on market participation by further reducing confidence and certainty to act and making individual efforts on climate action more isolated and less feasible. Second, it reduces the ability of actors to explore and access market opportunities on climate action that sit outside their priority outcomes. This presents the potential for a major barrier to critical enablers such as partnership development, investment and innovation. Third, it provides no platform for market actors to come together on common challenges to progress collective actions. In this situation the costs associated with addressing common issues are likely to be either: (i) avoided, because no one actor is solely responsible; (ii) inefficiently paid by actors multiple times across the economy; or (iii) ineffective in their impact compared to the scale of the problem. Finally, it increases the risk of those with the least power in the market being left behind, highlighting that market participation is a social justice issue, particularly for the vulnerable and marginalized (Wright and Nyberg, 2016).

Drawing on the above, Table 1 outlines eight traditional market failure indicators that highlight the persistence of a systemic market failure causing climate inaction.

## 3. Climate policies are missing the systemic nature of the challenge

We have identified four issues with climate policies that explain why they have failed to identify, consider and address climate inaction as a systemic market failure and development challenges.

First, many climate policies fail to consider inaction as an underlying problem and therefore a systemic economic and social issue preventing the transformation needed for sustainable development (Ross, 2016; Ford and Berrang-Ford, 2016). We see this in growing evidence of a divergence between the overly technical adaptation and mitigation focus of many climate policies and the broader need for climate action

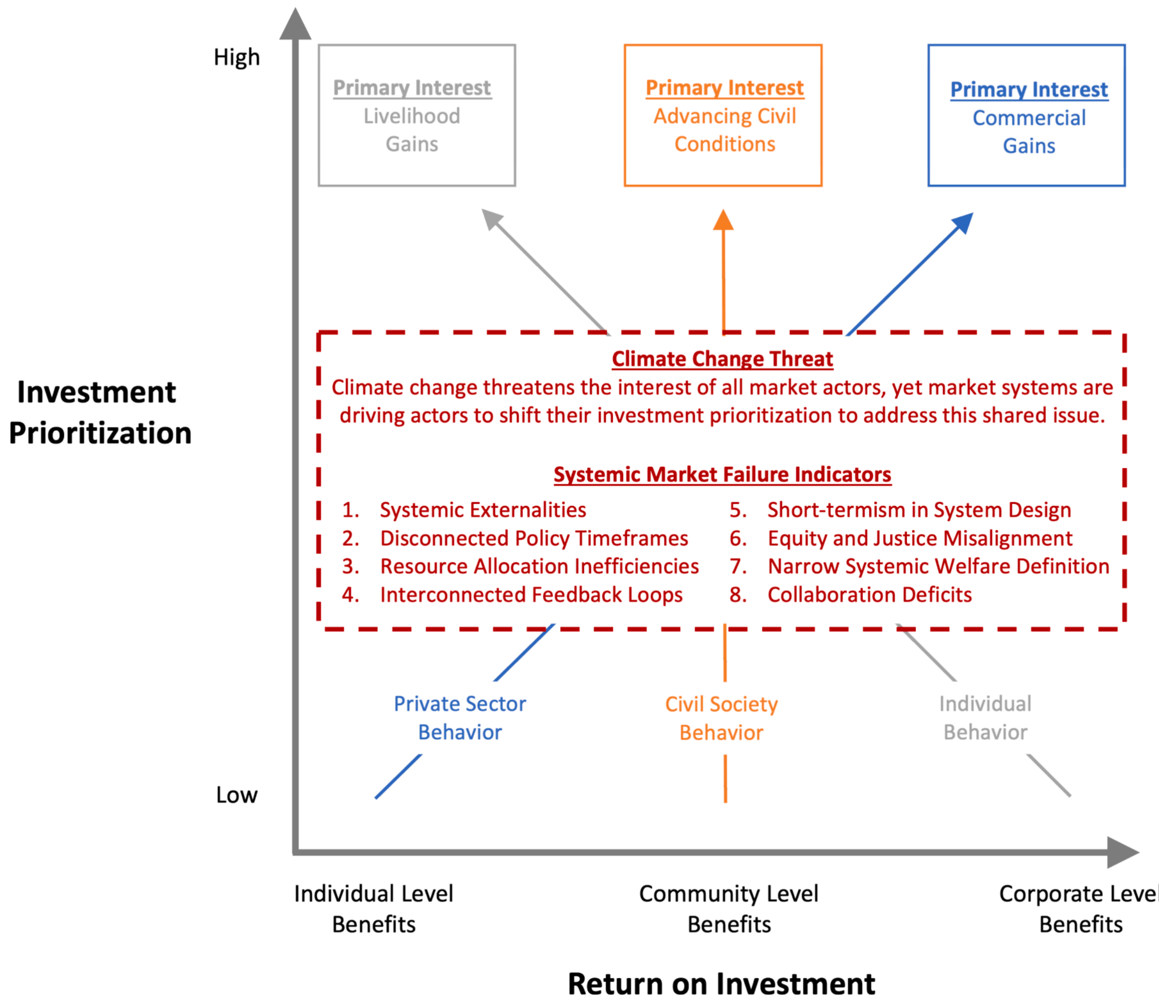


**Fig.1.** Key market actor behavior on climate action and the systemic market failure.

**Table 1**
Unpacks the indicators of systemic market failure causing climate inaction.

| |
|---|
| **Systemic Externalities**: In a connected system, not accounting for the full cost of emissions—like impacts on public health, ecosystems, and sea levels—leads to an overuse and depletion of natural resources. Systemically, this causes a chain reaction of environmental degradation, affecting multiple subsystems like health, agriculture, and water resources. |
| **Disconnected Policy Timeframes**: The system's resilience is challenged when mitigation focuses on long-term goals while adaptation targets immediate to medium-term objectives. This discrepancy weakens the overall effectiveness of climate change policies, leading to vulnerabilities in different subsystems. |
| **Resource Allocation Inefficiencies**: Siloed handling of climate policies creates competition for scarce resources in the system. Instead of pooling and optimally allocating resources to effectively address climate challenges, they are dispersed ineffectively, leading to systemic inefficiencies. |
| **Interconnected Feedback Loops:** The system's interconnectedness means that certain mitigation strategies might be influenced by adaptation choices and vice versa. A disruption in one part of the system might lead to unexpected consequences in another. |
| **Short-termism in System Design**: Prioritizing immediate gains over the system's long-term sustainability can restrict benefits from innovative, green solutions. Systemically, this results in lost opportunities for sustainable development, making future adaptations more challenging and costly. |
| **Equity and Justice Misalignment:** The system is not homogeneous. Developed parts of the system might focus on mitigation, side-lining urgent adaptation needs in developing regions. This misalignment can lead to increased vulnerabilities in certain subsystems, amplifying the impacts of climate change. |
| **Narrow Systemic Welfare Definition:** Overlooking broader values like biodiversity, cultural heritage, and indigenous rights in the system can create imbalances. Such a narrow focus can lead to degradation in non-economic subsystems, affecting overall system health and resilience. |
| **Collaboration Deficits**: Lack of interconnectivity between different parts of the system—whether intergovernmental or cross-sectoral—leads to fragmented responses. This issue can have a dynamic relationship of influence and impact with other drivers of systemic market failure, for example poor coordination can exacerbate resource misallocation – however this does not mean the cause of these is linked. The system requires coordinated efforts to address the multi-dimensional challenges of climate change effectively. |

that delivers broad-based sustainable development (Soergel, 2021; Fuso Nerini, 2019; Dupuis and Knoepfel, 2013; United Nations. Transforming Our World: The, 2030). The inclusion of climate action as an SDG expands the policy objective beyond mitigation and adaptation, to include driving an agenda of transformation that can result in sustainable economic and social development (Slawinski et al., 2017). From a development perspective, this framing carries real significance. It suggests that systemic issues exist that are hindering society from taking economy-wide climate action, and if left to existing markets and systems, climate action is unlikely to be effectively addressed or result in the best outcome for the economy and society (Jackson et al., 2021). While adaptation and mitigation are key outcomes sought by climate action, when considered as standalone policy instruments they haven't proven successful in enabling societies to overcome the systemic issues preventing sustainable development (Vogt-Schilb and Hallegatte, 2017). As Vogt-Schilb and Hallegatte (2017) highlight, climate policies should be designed so that they contribute to and strengthen the broader objectives and condition of the political economy (Kirby and O'Mahony, 2018).

Second, if sustainable development is the broader policy objective of climate action, the long-standing discipline and practice of development studies is significantly under-represented in the thinking behind many climate policies (Johnson, 2009). For instance, it is widely accepted and practiced in development economics that: (i) when effectively established, managed and accessible, markets have the greatest potential to deliver both social and economic development outcomes for society; and (ii) governments have a clear role in ensuring that the markets and systems (new or existing) are in place to enable sustainable development (Hallegatte et al., 2012). In stark contrast, the global landscape of climate action achieved under the last decade of climate policies and existing market systems is more reflective of a systemic market failure that policy interventions should seek to address. Confidence to fully invest in the political, financial, economic and social actions needed to align society on future value remains limited and the new market systems needed to deliver this change are not yet in place (Kruse et al., 2020; Oecd, 2021; Rickards et al., 2014). In addition, much needed collective action among sectors to create new value and drive systems change remains low and the successful upscaling of climate actions to a level where systems change and sustainable development occurs is well short of what is required (Den Uyl and Russel, 2018; Company, 2021). For example, despite increasing electric vehicle adoption, the transition to widespread EV use remains slow. A collective approach is lacking in many countries between automakers, energy providers, and governments to establish comprehensive charging infrastructure and incentive programs to accelerate adoption (Cullenward and Victor, 2020).

Third, many climate policies have tended to be neo-classical in their approach (Ferrannini et al., 2021), with limited use of industrial policy style interventions (Mandaroux et al., 2023). For example, The EU Emissions Trading Scheme primarily operates on a cap-and-trade basis and sets a price on carbon emissions and allows companies to buy and sell emission allowances. However, this neo-classical system has been criticized as being insufficient in fostering the necessary technological advancements and systemic changes needed for climate action (Waage et al., 2015). This neo-classical approach draws policy attention away from the systemic nature of inaction and focuses more on the isolated concerns of sectors, carbon markets and financial risks (Rosenbloom et al., 2020; Campiglio, 2018; Meckling and Allan, 2020). Under this approach, climate policies require only minimal and targeted policy interventions to correct the market (Ferrannini et al., 2021). In some situations, this may be true, such as the need for regulatory mechanisms that support corporations to make climate-related financial disclosures. However, the level of inaction to date highlights that the sustainable development pathway ahead requires a more systemic change to market-based economies. This perhaps explains the re-emergence of industrial policy over the past decade as an alternative approach to guide and support climate action and sustainable development in market-based economies (Noll and S. B. and S. T. S. Domestic-First, , 2024). In its reinvigorated form, industrial policy presents the case for significant government intervention where systemic market failures inhibit development. It argues that governments have a unique role, obligation and capability to help societies address complex development issues like climate inaction (Mandaroux et al., 2023). Early examples of industrial policy for climate action are emerging, such as the US Inflation Reduction Act (Levinson et al., 2024), which seeks to mobilize US $5.6 trillion in global benefit for climate action by 2050 (Peng, 2021). However, many governments are still yet to take up their unique role in being able to reorient economic and social systems and fully explore industrial style climate policies as an approach to overcoming systemic development issues like climate inaction (Vogt-Schilb and Hallegatte, 2017).

Fourth, climate policies often skip past people, communities and organizations as necessary actors and rational optimizers and focus on technical areas of science, technology and communication (Susskind, 2022). This is evident in the historical opposition many local communities in the US had to renewable energy projects, which slowed the energy transition and is estimated to have cost that country 4600 MW between 2008 and 2021 (Jagers, 2020). This oversimplification of behaviors misses the need to activate people, communities and organizations as critical actors that can deliver change in society (Bouman et al., 2021), which requires greater consideration of social and personal norms and values (United Nations and I. T. F. on F , 2021. (2021).). This is precisely why development finance and assistance in emerging markets over many decades has routinely coupled investment in technical areas with support for capacity building among people, communities and organizations as key actors of change (Reser and Bradley, 2020). It is becoming increasingly evident that successful climate action will likely be dependent on supporting societal engagement, participation, and behavioral change (Ruffer et al., 2018), which are all key for systems change and underlying conditions for development (Jackson et al., 2021).

## 4. Reframing climate inaction as a development challenge

Many of the elements needed to consider climate inaction as a development challenge already exist but need clarity in the form of a coherent policy framework.

*What is the policy objective?*

Using a sustainable development lens, the focus of climate policies should be on addressing climate inaction as a systemic market failure preventing sustainable development. This necessitates transformative policy that can build societies' capability to overcome systemic issues, such as those causing climate inaction. Underpinning this is the unique role of governments to enable sustainable development by driving both market and systems change, which is a fundamental principle of development. Therefore, a policy seeking to address climate inaction as a systemic market failure and development challenge would require government intervention capable of (i) enabling the market conditions needed for society to address climate inaction as a risk to sustainable development; and (ii) where necessary, ensuring public institutions have the capability and legitimacy to help society address climate inaction in the most democratic, equitable and efficient way (Ray, 1998; Hallegatte et al., 2012).

*What market conditions are required?*

In declaring climate change the greatest market failure and threat to development, the Stern review argued the case for climate change to be valued throughout the entire economy and action supported across all sectors of society (Stern, 2007). In doing so, it presented considerable detail outlining the market conditions required to advance transformative climate action and avoid the impacts of climate inaction. We have drawn three overarching market conditions from the Stern Review that simplify what is required from climate policies to address climate inaction as a systemic market failure. The first is the need for policies to generate enhanced participation and collective action between key actors that wouldn't otherwise happen via markets, including the public and private sectors, civil society and individuals, which we define as building the market ecosystem. The second is the need to motivate a change in understanding, behavior and investment patterns of actors, which we define as providing market enablers. The third is the need for institutional safeguards that provide certainty and confidence in the operating environment, which we define as providing market assurances.

*What should guide government interventions?*

Where public institutions have sufficient capability, there are established development approaches designed specifically for guiding interventions to overcome systemic failures in markets. A 'market systems' or 'market systems development' (MSD) approach is best positioned to guide the necessary government interventions here. MSD is widely applied in development finance and assistance to address a variety of systemic development challenges, including everything from broad economic issues to sectoral and social issues (Bird, 2024). While no evidence exists of a MSD approach being deployed specifically to address widespread climate inaction, it is emerging as a possible solution for important climate action issues such as just transition, as well as sector specific actions like climate-smart agriculture (Dfat, 2020). What makes MSD particularly relevant for addressing climate inaction is a shared objective to address systemic issues that prevent economic and social development (Nutz, 2017; Centre and Guide, 2015). Unlike typical market interventions that focus on small corrections of existing markets, MSD targets the development of market systems that reflect the value associated with society's longer-term needs. In doing so, MSD also seeks to enhance equity and inclusion (Osorio-Cortes et al., 2021), which are critical aspects of climate justice and realizing a just transition (Wright and Nyberg, 2016). The system focus of MSD makes it capable of overcoming systemic challenges like climate inaction. There is a considerable body of grey literature on MSD to draw from, which highlights that when effectively designed and targeted towards systemic issues and behavioral change, MSD interventions can be highly effective (Kramer and Pfitzer, 2016).

## 5. A market systems framework for guiding climate policies on inaction

To address climate inaction as a systemic market failure, interventions that take a systems view of markets are needed. Fig. 2 proposes the key components of a market system framework that can guide such interventions. It combines standardized components of the widely implemented MSD approach with the three overarching market conditions for addressing climate change drawn from the Stern Review. Like all market approaches, the core of the market system in this framework is supply and demand. The market ecosystem presented highlights the need to ensure that all market actors (public sector, private sector, civil society and individuals) have clear opportunities for market access. This includes ensuring a pathway for direct access, where actors should individually participate in climate action; and partnership, where actors should be taking collective action. The market enablers and assurances presented mirror the supporting functions and rules outlined in a standard MSD framework. These represent the areas where well-designed and mutually reinforcing interventions should be targeted to improve

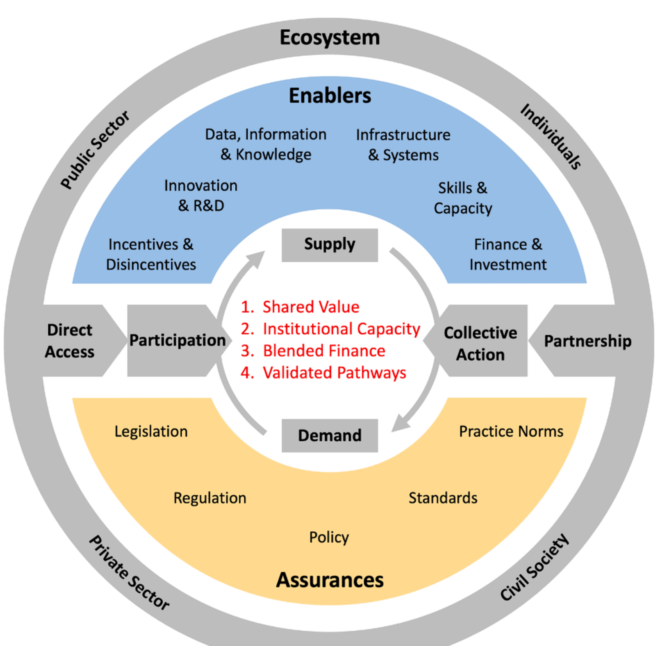


**Fig.2.** A market systems framework that contains the key components needed to address climate inaction as a systemic market failure.

market conditions and help overcome climate inaction.

For enablers, this includes ensuring targeted incentives and disincentives, innovation and R&D, knowledge and information, infrastructure and systems, skills and capacity, and finance and investment. For assurances, this includes providing clarity and stability needed for the market to act through legislation, regulation, policy, standards and culture. At the centre of the framework are the principles to break the deadlock of climate inaction as a result of systemic market failure. These principles and their justification are provided below.

**Principle 1**. (*Facilitate market systems that clarify and enable shared value on climate action*) *The systemic nature of climate change means that overcoming climate inaction necessitates all market actors to convene around shared challenges across the economy and society. To realize this, it is critical that climate policies work across sectors and communities to establish a clear understanding of common climate risks, opportunities, dependencies and priorities. Although diversity among actors does present complexity in advancing this, coalescing this diversity is undoubtedly key to unlocking the shared value needed to overcome climate inaction and drive collective action. This can be achieved by climate policies facilitating an ecosystem where collective impact can be achieved among actors* (Flood et al., 2015). *The focus here should be on enabling* (1) *a shared agenda*, (2) *continuous communication*, (3) *mutually reinforcing activities*, (4) *a shared measurement system and* (5) *backbone support*(Bryson et al., 2015).

To manage the inevitable complexities that will arise from facilitating shared value on climate action, climate policies will need to ensure transparency and equity among actors throughout the process. A network governance model is worth considering here (Bodin, 1979). With the right leadership network governance models have shown that they can build and oversee collaboration on common issues across complex socio-ecological systems (Howes, 2015). There are also some advancements in applying this model to address common climate change problems that need to be noted. These align well with facilitating collective impact and highlight the importance of (1) developing a shared policy vision, (2) adopting multi-level planning, (3) integrating with legislation (4) networking organizations, and (5) establishing cooperative funding (Tosun and Schoenefeld, 2017). Above all, network governance arrangements must be supported by strong public institutions and uphold principles of legitimacy, integrity and equity (Lawrence et al., 2022). By facilitating market systems that encourage shared value, businesses, investors, and consumers can align economic incentives with climate-positive actions. This ensures that actors in the market are not just driven by profit, but also by the shared value of climate risk reduction and incorporates externalities through a shared valuation.

**Principle 2**. (*Strengthen the capacity of our public institutions as key actors in climate action*) *While there is a case for public institutions intervening on systemic issues like climate inaction or indeed more actively managing the climate risks they face, the capability to do either is often limited. Chapters of the Sixth Assessment Reports of the Intergovernmental Panel on Climate Change (IPCC) describe the inability of governments and institutions to deal with the increasing scale and scope of climate change as a new type of climate risk* (Boin, 2016). *This situation highlights that public institutions themselves are not immune to climate inaction and necessitates the need to advance capacity building and institutional strengthening on climate action in many public institutions* (Reser and Bradley, 2020). *Unfortunately; the knowledge base to guide this work is limited*(Runhaar et al., 2017); *which is an issue requiring urgent attention. However; even when relevant knowledge is available, public servants still struggle to apply concepts like climate governance, climate risk and mainstreaming within standard institutional policies, planning, systems and proceedures* (Greene, 2018). *As key actors in climate action; public institutions must have sufficient support to rapidly overcome any knowledge, practice and institutional capacity challenges they face. Strengthening public institutions ensures that they are adequately equipped to lead, coordinate, and enforce climate action policies. This can lead to clearer* *policy direction, more efficient use of resources, and increased collaboration among different sectors, thereby addressing systemic barriers to climate action. Failing to do so is likely to weigh negatively on the market system, result in less desirable market conditions, allow inaction to persist and ultimately threaten sustainable development.*

Given a key objective of development is to ensure public institutions have the capability to help society address market failures, many development insights are transferable here. Foremost is the need to establish longer-term support and technical assistance arrangements that can help traverse all aspects of an institution and objectively facilitate institutional strengthening and capacity building. Partnerships with trusted academic, expert and civil society entities are essential here to ensure effective strategy, skills development and implementation (Dany et al., 2015). With the right supports in place, it should then be possible to assess a public institution's existing capacity for climate action more effectively and identify concrete steps for how this can be improved (Wise, 2014). Where possible such assessments should be driven by public institutions to develop ownership and capacity throughout the process (Biesbroek et al., 2018). It is also recommended that any assessment process determines and aligns with administrative traditions (van Vliet et al., 2015), which are the values, structures and functions that guide a public institution. Finally, in the absence of well-established knowledge and frameworks, there is an obligation on public institutions to allow for learning throughout their capacity-building and institutional strengthening processes on climate action. The implementation nature of action research can guide here and several case studies exist that demonstrate how to apply action research on climate action in public institutions (Pedley, 2010).

**Principle 3**. (*Ensure finance blending structures exist that enable investment in climate action*) *Realizing climate action requires collaborative financing structures. Blended financing structures are increasingly applied to attract the investment needed to design, finance and deliver better outcomes on issues such as to leverage climate action. Finance blending structures combine public and private investments, leverages the strengths of both public and private finance, creating a more robust financial foundation for climate initiatives, and ensuring that capital flows to where it's most needed.*

These financing approaches have evolved out of the international development sector as a way to leverage public investment to attract private co-investment for overcoming common development issues. The aim is to achieve this by establishing a financing structure that can deliver, and even optimize for, social, environmental and economic outcomes that benefit both investors and beneficiaries. This is usually in the form of a pooled fund or facility, that administers standard investment vehicle types like shares, bonds, secured debt or tax-exempt grants (Choi and Seiger, 2020). From an investment perspective, the advantage of blended finance is that many collective investment vehicles are already well understood and regulated within commercial finance, increasing the ease and lowering the risk of private participation. In addition, public funds can provide credit enhancement, send a strong market signal and crowd in more and different investors and capital. This works best when public spending is targeted to mobilize the maximum amount of private capital co-investment through targeted and strategic investments and alignment of incentives. When well-designed, blended finance structures can meet the needs of different types of investors with different risk and return requirements and enable capital to be directed in ways that may not otherwise be possible.

Since 2009, it is estimated that blended finance structures have leveraged over USD140 billion of private financing for international development (Andersen, 2019). The OECD is now pushing blended finance as a possible financing solution for SDGs (Tonkonogy et al., 2018), with early signs of commercial demand for investment in both clean energy (Alliance, 2024) and general climate financing (Andersen, 2019) being positive. In the context of climate inaction and the need to mobilize large volumes of investment to drive rapid climate action, blended finance stands out as an established financing model worthy of uptake by climate policies. Key design considerations include understanding the key actors to be mobilized, structural barriers to investment, information asymmetry, uncertainty constraining market development and where concessional capital can assist in overcoming a short-term lack of track record. Many of these tools can be adapted from broader economic policy designed to stimulate growth and innovation.

Green finance plays a crucial role in shaping sustainable development within future market systems, however some basic gaps still exist. The recent Glasgow COP26 Financial Alliance for Net Zero Progress Report 2024 highlighted the need to address the data gap, action gap and investment gap among financial institutions, in order to unlock finance for climate action (Choi and Seiger, 2020). Blended finance can be a tool to help overcome these gaps by reducing transaction cost and providing clarity on early market investment priorities. Ensuring strong governance and assurance of green and blended finance structures can provide critical leadership and confidence to grow and deploy much needed green finance for climate action (Lamperti et al., 2019).

**Principle 4**. (*Ensure that market system interventions are effectively modeled and validated*) *The enablers and assurances outlined in* Fig. 2 *present the common levers available to policymakers seeking to intervene in climate inaction.* Princile 1 *preempts that successful market system strengthening will in part be dependent on a representative sense of shared value on climate action from actors across society and the economy. However, for an intervention to be effective, policymakers need to go further than just activating the available levers and shared value. Where possible climate policies need to model and validate potential market system interventions to determine the most effective pathways for driving action and sustainable development*(Stern, 2016). *This is a critical aspect of market system development; yet often overlooked or poorly applied by climate policies. Effective modeling and validation ensure that market interventions are based on reliable data, sound science, and thorough analysis. By forecasting the potential impacts of interventions, policymakers can refine strategies, mitigate negative outcomes, and amplify positive shared impacts. This fosters trust in the interventions and ensures resources are used efficiently. Essentially, climate policies need to ensure the real economic costs and benefits of a potential intervention along with its assumptions are properly understood and can be refined where necessary* (Babatunde et al., 2017).

Although still an evolving area, there is increasing evidence to suggest that a modelling toolkit is available for policymakers to consider the cost and benefits of interventions. In particular, a combination of dynamic stochastic computable general equilibrium (DSGE) models and agent-based models (ABMs) is forming as the preferred approach (Lamperti et al., 2018). What makes a hybrid use of DSGE and ABM so important for policymakers is the ability to consider the value of interventions on inaction based on both macroeconomic changes over time and the diverse socio-economic behaviors of market actors (Castro, 2020). The micro focus of ABM modelling also helps enable the consideration of a variety of interventions (Lamperti et al., 2019).

## 6. Conclusion

Climate inaction remains a major threat to sustainable development, despite decades of climate policies and growing evidence of the costs of inaction outweighing the costs of action. We argue that current climate policies often fail to elevate and translate their adaptation and mitigation objectives into broader economic impact and sustainable development. We identified climate inaction as a systemic market failure that is preventing transformative climate action across market-based economies. This systemic failure is characterized by isolated non-state actor activity, limited collective action, and minimal resources being directed towards climate action

Four key issues with climate policies are identified that explain why they have failed to address climate inaction as a systemic challenge. First, policies fail to consider inaction as an underlying problem and systemic economic issue preventing sustainable development transformation. Second, development studies and economics are under-represented in climate policy thinking. Third, climate policies have tended to be neo-classical in approach with limited use of industrial policy-style interventions to drive systems change. Lastly, policies often skip past people, communities and organizations as critical actors and focus narrowly on technical areas.

To reframe climate inaction as a development challenge, the policy objective should be to build society's capability to overcome systemic issues preventing sustainable development. This requires government interventions guided by a market systems development approach to 1) build the market ecosystem for enhanced participation and collective action 2) provide market enablers to motivate behavior change and investment, and 3) ensure market assurances provide certainty and confidence.

Addressing climate inaction as a systemic market failure and development challenge demands a stronger role for governments in driving both market and systems change. With the climate emission clock ticking, policymakers and scholars must focus more attention on this critical issue.

## CRediT authorship contribution statement

**Samuel Mackay:** . **Rob Hales:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **John Hewson:** Writing – review & editing, Writing – original draft, Conceptualization. **Rosemary Addis:** Writing – review & editing. **Brendan Mackey:** Writing – review & editing, Writing – original draft, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

No data was used for the research described in the article.

## References


Andersen, O.W., et al., 2019. *Blended Finance Evaluation: Governance and Methodological Challenges*. https://doi.org/10.1787/4c1fc76e-en.

Andrew, B. Market failure, government failure and externalities in climate change mitigation: The case for a carbon tax. in *Public Administration and Development* vol. 28 (2008).

Babatunde, K.A., Begum, R.A., Said, F.F., 2017. Application of computable general equilibrium (CGE) to climate change mitigation policy: A systematic review. *Renew. Sustain. Energy Rev.* 78, 61–71.

Bäckstrand, K., Kuyper, J. W., Linnér, B. O. & Lövbrand, E. Non-state actors in global climate governance: from Copenhagen to Paris and beyond. *Environmental Politics* vol. 26 Preprint at https://doi.org/10.1080/09644016.2017.1327485 (2017).

Biermann, F., et al., 2022. Scientific evidence on the political impact of the Sustainable Development Goals. *Nat Sustain.* https://doi.org/10.1038/s41893-022-00909-5.

Biesbroek, R., Peters, B.G., Tosun, J., 2018. Public Bureaucracy and Climate Change Adaptation. *Rev. Policy Res.* 35, 776–791.

Bird, J. *Market Systems Development for a Just Transition: What Does It Really Mean for Donors and Practitioners?* https://www.ilo.org/sites/default/files/2024-06/MSD%20and%20JT%20-%20v1.1.pdf (2024).

Bodin, Ö. Collaborative environmental governance: Achieving collective action in social-ecological systems. *Science (1979)* **357**, (2017).

Boin, Arjen, and M. L. Designing resilient institutions for transboundary crisis management: A time for public administration. *Public Adm* **94**, 289–298 (2016).

Bouman, T., Steg, L., Dietz, T., 2021. Insights from early COVID-19 responses about promoting sustainable action. *Nat Sustain* 4, 194–200.

Bromley, D.W., 2007. Environmental regulations and the problem of sustainability: Moving beyond 'market failure'. *Ecol. Econ.* 63.

Bryson, J.M., Crosby, B.C., Stone, M.M., 2015. Designing and Implementing Cross-Sector Collaborations: Needed and Challenging. *Public Adm Rev* 75, 647–663.

Campiglio, E., et al., 2018. Climate change challenges for central banks and financial regulators. *Nat Clim Chang* 8, 462–468.

Castro, J., et al., 2020. A review of agent-based modeling of climate-energy policy. *Wiley Interdiscip Rev Clim Change* 11, 1–26.

The Springfield Centre. *The Operational Guide for the Making Markets Work for the Poor (M4P) Approach*. (2015).

Choi, E., Seiger, A., 2020. Catalyzing Capital for the Transition toward Decarbonization : Blended Finance and Its Way Forward Acknowledgment. *SSRN* 1–42.

McKinsey & Company. *Voices on Infrastructure: Scaling EV Infrastructure to Meet Net-Zero Targets*. (2021).

Cullenward, D., Victor, D.G., 2020. *Making Climate Policy Work*. John Wiley & Sons.

Dany, V., Bowen, K.J., Miller, F., 2015. Assessing the institutional capacity to adapt to climate change: a case study in the Cambodian health and water sectors. *Clim. Pol.* 15, 388–409.

Den Uyl, R.M., Russel, D.J., 2018. Climate adaptation in fragmented governance settings: the consequences of reform in public administration. *Env Polit* 27, 341–361.

Derman, B.B., 2014. Climate governance, justice, and transnational civil society. *Clim. Pol.* 14, 23–41.

Dfat, 2020. *Synthesis Review of DFAT-Funded Market Systems Development Initiatives. Part I. Main. Report*.

Dupuis, J., Knoepfel, P., 2013. The adaptation policy paradox: The implementation deficit of policies framed as climate change adaptation. *Ecol. Soc.* 18.

Ferrannini, A., Barbieri, E., Biggeri, M., Di Tommaso, M.R., 2021. Industrial policy for sustainable human development in the post-Covid19 era. *World Dev* 137, 105215.

Flood, J., Minkler, M., Hennessey Lavery, S., Estrada, J., Falbe, J., 2015. The Collective Impact Model and Its Potential for Health Promotion: Overview and Case Study of a Healthy Retail Initiative in San Francisco. *Health Educ. Behav.* 42, 654–668.

Ford, J.D., Berrang-Ford, L., 2016. The 4Cs of adaptation tracking: consistency, comparability, comprehensiveness, coherency. *Mitig Adapt Strateg Glob Chang* 21, 839–859.

Fuso Nerini, F., et al., 2019. Connecting climate action with other Sustainable Development Goals. *Nat Sustain* 2, 674–680.

Glasgow Financial Alliance for Net Zero (GFANZ). *GFANZ 2024 Progress Report*. (2024).

Clare Shakya Naman Gupta Zac Bull Sam Greene, K. C. *Building Institutional Capacity for Enhancing Resilience to Climate Change: An Operational Framework and Insights from Practice*. (2018).

Hallegatte, S., Shah, A., Brown, C., Lempert, R., Gill, S., 2012. *Investment Decision Making Under Deep Uncertainty – Application to Climate Change. World Bank Policy Research Working Paper*.

Hornsey, M.J., Fielding, K.S., 2020. Understanding (and Reducing) Inaction on Climate Change. *Soc Issues Policy Rev* 14, 3–35.

Howes, M., et al., 2015. Towards networked governance: improving interagency communication and collaboration for disaster risk management and climate change adaptation in Australia. *J. Environ. Plan. Manag.* 58, 757–776.

Hsu, A., Brandt, J., Widerberg, O., Chan, S., Weinfurter, A., 2020. Exploring links between national climate strategies and non-state and subnational climate action in nationally determined contributions (NDCs). *Clim. Pol.* 20, 443–457.

Jackson E.A., J. M. Understanding Market Failure in the Developing Country Context. in *Decent Work and Economic Growth. Encyclopedia of the UN Sustainable Development Goals* (ed. Leal Filho W., Azul A.M., Brandli L., Lange Salvia A., W. T.) (Springer, Cham., 2021). doi:https://doi.org/10.1007/978-3-319-95867-5_44.

Jagers, S.C., et al., 2020. On the preconditions for large-scale collective action. *Ambio* 49, 1282–1296.

Johnson, S. Institutions, Markets and Economic Development. in *An Introduction to the Human Development and Capability Approach* 184–196 (Routledge, 2009). doi: 10.4324/9781849770026-28.

Kahn, M., et al., 2019. Long-Term Macroeconomic Effects of Climate Change. *IMF Working Papers* 19.

Kirby, P. & O'Mahony, T. Development Models: Lessons from International Development. in *The Political Economy of the Low-Carbon Transition* 89–113 (Springer International Publishing, Cham, 2018). doi:10.1007/978-3-319-62554-6_4.

Kramer, M.R., Pfitzer, M.W., 2016. The ecosystem of shared value. *Harv Bus Rev* 2016.

Kruse, T., Mohnen, M., Sato, M., 2020. *Are Financial Markets Aligned with Climate Action? New Evidence from the Paris Agreement. Centre for Climate Change Economics and Policy Working Paper 364/Grantham Research Institute on Climate Change and the Environment Working Paper 333*.

Lamperti, F., Dosi, G., Napoletano, M., Roventini, A., Sapio, A., 2018. Faraway, So Close: Coupled Climate and Economic Dynamics in an Agent-based Integrated Assessment Model. *Ecol. Econ.* 150, 315–339.

Lamperti, F., Mandel, A., Napoletano, M., Sapio, A., Roventini, A., Balint, T., & Khorenzhenko, I. Lamperti, F., et al., 2019. Towards agent-based integrated assessment models: examples, challenges, and future developments. Reg Environ Change 19, 747–762.

Lawrence, J., B. Mackey, F. Chiew, M.J. Costello, K. Hennessy, N. Lansbury, U.B. Nidumolu, G. Pecl, L. Rickards, N. Tapper, A. Woodward, and A. W. Australasia. in *In:Climate Change 2022: Impacts, Adaptation, and Vulnerability.Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (ed. H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. R.) (Cambridge University Press. In Press., 2022).

Arik Levinson, Karl Dunkle Werner, Matthew Ashenfarb & Annelise Britten. *The Inflation Reduction Act's Benefits and Costs*. (2024).

Linnenluecke, M.K., Griffiths, A., Winn, M.I., 2013. Firm and industry adaptation to climate change: a review of climate adaptation studies in the business and management field. *Wiley Interdiscip Rev Clim Change* 4, 397–416.

Mandaroux, R., Schindelhauer, K., Basse Mama, H., 2023. How to reinforce the effectiveness of the EU emissions trading system in stimulating low-carbon technological change? Taking stock and future directions. *Energy Policy* 181.

Meckling, J., Allan, B.B., 2020. The evolution of ideas in global climate policy. *Nat Clim Chang* 10, 434–438.

Newman, R., Noy, I., 2023. The global costs of extreme weather that are attributable to climate change. *Nat Commun* 14.
Noll, B. , S. B. and S. T. S. Domestic-First, Climate Second? Global Consequences of the Inflation Reduction Act. *Joule* **8**, 1869–1873 (2024).
Nutz, N. *A Guide to Market-Based Livelihood Interventions for Refugees*. (2017).
Oecd, 2021. *Financial Markets and Climate Transition: Opportunities. Challenges and Policy Implications.*
Osorio-Cortes, L. & Albu, M. The Results Achieved by Programmes Using a Market Systems Approach: Narrative Synthesis of Current Evidence. (2021).
Pedley, P., 2010. *Blended Finance: What It Is and How It Works*. https://doi.org/10.4324/9780203403839_chapter_1.
Peng, W., et al., 2021. Climate policy models need to get real about people - here's how. *Nature* 594, 174–176.
Ray, D., 1998. *Development Economics*. Princeton University Press.
Reser, J.P., Bradley, G.L., 2020. The nature, significance, and influence of perceived personal experience of climate change. *Wiley Interdiscip Rev Clim Change* 11, 1–28.
Rickards, L., Wiseman, J., Kashima, Y., 2014. Barriers to effective climate change mitigation: The case of senior government and business decision makers. *Wiley Interdiscip Rev Clim Change* 5, 753–773.
Romanello, M., et al., 2023. The 2023 report of the Lancet Countdown on health and climate change: the imperative for a health-centred response in a world facing irreversible harms. *Lancet*.
Rosenbloom, D., Markard, J., Geels, F.W., Fuenfschilling, L., 2020. Why carbon pricing is not sufficient to mitigate climate change—and how "sustainability transition policy" can help. *Proc Natl Acad Sci U S A* 117, 8664–8668.
Ross, L., et al., 2016. The Climate Change Challenge and Barriers to the Exercise of Foresight Intelligence. *Bioscience* 66, 363–370.
Ruffer, T., Bailey, H., Dahlgren, S., Spaven, P., Winters, M., 2018. *Evaluation of the Market Systems Development Approach. Lessons for Expanded Use and Adaptive Management at Sida. I*.
Runhaar, H., Wilk, B., Persson, Å., Uittenbroek, C., Wamsler, C., 2017. Mainstreaming climate adaptation: taking stock about "what works" from empirical research worldwide. *Reg Environ Change*. https://doi.org/10.1007/s10113-017-1259-5.
Sachs, J., Kroll, C., Lafortune, G., Fuller, G., Woelm, F., 2021. *Sustainable Development Report 2021*. *Sustainable Development Report* 2021. https://doi.org/10.1017/9781009106559.
Sanderson, B.M., O'Neill, B.C., 2020. Assessing the costs of historical inaction on climate change. *Sci Rep* 10, 1–12.
Skovgaard, J., van Asselt, H., 2019. The politics of fossil fuel subsidies and their reform: Implications for climate change mitigation. *Wiley Interdiscip Rev Clim Change* 10, 1–12.
Slawinski, N., Pinkse, J., Busch, T., Banerjee, S.B., 2017. The Role of Short-Termism and Uncertainty Avoidance in Organizational Inaction on Climate Change. *Bus Soc* 56, 253–282.
Soergel, B., et al., 2021. A sustainable development pathway for climate action within the UN 2030 Agenda. *Nat Clim Chang* 11.
Stern, N., 2007. The economics of climate change: The stern review. *The Economics of Climate Change: the Stern Review* 9780521877, 1–692.
Stern, N., 2016. Economics: Current climate models are grossly misleading. *Nature* 530, 407–409.
Stern, N., Stiglitz, J., Taylor, C., 2022. The economics of immense risk, urgent action and radical change: towards new approaches to the economics of climate change. *J. Econ. Methodol.* 29, 181–216.
Susskind, L., et al., 2022. Sources of opposition to renewable energy projects in the United States. *Energy Policy* 165.
Thøgersen, J., 2021. Consumer behavior and climate change: consumers need considerable assistance. *Curr Opin Behav Sci* 42, 9–14.
Tollefson, J., 2021. Top climate scientists are sceptical that nations will rein in global warming. *Nature* 599, 22–24.
Tonkonogy, B., Brown, J., Micale, V., Wang, X. & Clark, A. Blended Finance in clean energy. (2018).
Tosun, J., Schoenefeld, J.J., 2017. Collective climate action and networked climate governance. *Wiley Interdiscip Rev Clim Change* 8.
United Nations, 2020. *Emissions Gap Emissions Gap Report* 2020.
United Nations, I. T. F. on F. for D. *Financing for Sustainable Development Report 2021*. (2021).
United Nations. *Transforming Our World: The 2030 Agenda for Sustainable Development. A/RES/70/1* (2015) doi:10.1201/b20466-7.
van Vliet, M., van Buuren, A. & Eshuis, J. Action research for the governance of adaptation to climate change: Conclusions and lessons learnt. in *Action Research for Climate Change Adaptation: Developing and applying knowledge for governance* (eds. van Buuren, A., Eshuis, J. & van Vliet, M.) 182–196 (Routledge, London, 2015).
Vogt-Schilb, A. & Hallegatte, S. Climate policies and nationally determined contributions: reconciling the needed ambition with the political economy. *Wiley Interdisciplinary Reviews: Energy and Environment* vol. 6 Preprint at https://doi.org/10.1002/wene.256 (2017).
Waage, J, Yap, C, Bell, S, Levy, C, Mace, G, Pegram, T, Unterhalter, E, Dasandi, N, Hudson, D, Kock, R, Mayhew, S. H, Marx, C, and Poole, N. Governing Sustainable Development Goals: interactions, infrastructures, and institutions. in *Thinking Beyond Sectors for Sustainable Development* (ed. Waage, J and Yap, C.) Pp. 79–88 (Ubiquity Press, London, 2015). doi:https://doi.org/10.5334/bao.i.
Wise, R.M., et al., 2014. Reconceptualising adaptation to climate change as part of pathways of change and response. *Glob. Environ. Chang.* 28, 325–336.
Woodruff, S.C., Stults, M., 2016. Numerous strategies but limited implementation guidance in US local adaptation plans. *Nat Clim Chang* 6, 796–802.
Wright, C. & Nyberg, D. Climate change and social innovation. in *Handbook of Inclusive Innovation: The Role of Organizations, Markets and Communities in Social Innovation* (ed. George, Gerard. Baker, Ted. Tracey, Paul. Joshi, H.) 47–60 (Edward Elgar Publishing, Cheltenham, UK;Northampton, MA, 2016). doi:https://doi.org/10.4337/9781786436016.